Closed Forms for Platonic Helices

Exact parameters and an audit of the 28-species catalog

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Abstract

Repeated identical face-to-face placements of a rigid solid generate a screw motion, and the centroids of successive joining faces trace a segmented helix. For joint centers lying on equal-radius face normals, we express the principal subunit rotation, the angle between a segment and the oriented screw axis, the axial advance, and the joint-centroid radius in terms of three intrinsic inputs: the normal-alignment angle, the joining twist, and the distance between joint centers. A quaternion composition supplies a common denominator for all four parameters. We specialize the result to the five Platonic solids and give exact expressions for every entry in the previously published 28-species catalog, with edge length one. An audit distinguishes the rotation of a subunit from the geometry of a straight joint orbit, identifies the constructor-dependent length normalization of the earlier table, and derives exact replacements for three icosahedral twists that only approximately match face vertices. The original and corrected configurations are both tabulated. Symbolic identities and independent matrix, circumcircle, and vertex-matching checks accompany the derivations.

1 Introduction

Identical rigid subunits conjoined by the same rule form a periodic chain. The resulting motion is a screw motion, so a chosen point of a subunit traces a discrete helical orbit. This observation connects the geometry of protein helices, polyhedral chains, and engineered assemblies [2, 3, 1]. Exact descriptions of the tetrahelix and modified tetrahedral chains already exist, including exact rotation, radius, and pitch calculations [4, 5]. The question addressed here is how to obtain a uniform description from the geometry of the joining faces.

Read's 2022 paper supplied algorithms and a numerical catalog of 28 representative Platonic helices, and proposed closed-form descriptions as future work [1]. This paper develops such a description for the *helix through the joint-face centroids*. A vertex orbit or the orbit of the center of the solid can have a different radius under the same screw motion. Identifying the point whose orbit is being measured is therefore essential when comparing results.

The main result uses the intrinsic inputs

$$L = \|C - B\|, \qquad \alpha = \arccos(-\mathbf{n}_B \cdot \mathbf{n}_C), \qquad \tau,$$

where B, C are selected joint centers within one subunit, $\mathbf{n}_B, \mathbf{n}_C$ are their outward unit face normals, and τ is a twist measured from a specified normal-alignment reference. The angle α is generally *not*

the angle between the two outward normals. Nor should α, τ be identified without a derivation with the bond angle and torsion angle of a polygonal chain used in other helix descriptions [3].

The theorem applies when the centers, relative to a common origin, satisfy $B = \rho \mathbf{n}_B$ and $C = \rho \mathbf{n}_C$ for the same $\rho > 0$. Every Platonic solid has this property when its face centroids are used. Under this hypothesis,

$$Q = 1 - \cos^2(\alpha/2)\cos^2(\tau/2)$$

governs all four requested parameters. The rotation formula has a broader scope than the radius and advance formulas: face positions that do not satisfy the equal-radius hypothesis require additional positional data.

Two distinctions emerge when the closed forms are compared with the earlier catalog. First, a straight sequence of joint centers does not reveal the rotation of the solid about that line. Second, the published lengths agree with the original graphics constructors, which use unit circumradius for four solids and unit edge length for the cube. We give an explicit conversion to unit edges. A separate vertex audit finds that the three catalog twists $45°, -75°, 165°$ for one icosahedral face class are approximate matching choices. Their exact parameters remain well defined, and we retain them in the literal catalog tables. We also derive the exact matching twists and their parameters.

The word "species" below refers to the representatives and names inherited from the 2022 catalog. We do not infer uniqueness of an ordered discrete helix from its radius and axial advance alone: its angular step also matters. We also make no claim that every chain in the catalog avoids self-intersection.

2 Geometry, motion, and conventions

2.1 A periodic joint-centroid orbit

Place a reference solid with its center at the origin and choose two distinct faces. Let their centroids be B, C and their outward unit normals be $\mathbf{n}_B, \mathbf{n}_C$. For a Platonic solid of inradius ρ,

$$B = \rho \mathbf{n}_B, \qquad C = \rho \mathbf{n}_C. \tag{1}$$

Here $L = \|C - B\|$ is a straight-line distance within one solid. It is also the length of one segment of the joint-centroid helix. It is not the edge length, the circumradius, or the advance along the screw axis.

Let R be the forward rotation that makes the incoming B face of the next solid flush against the current C face. Thus $R\mathbf{n}_B = -\mathbf{n}_C$. The rigid placement is

$$T(\mathbf{x}) = R(\mathbf{x} - B) + C = R\mathbf{x} + \mathbf{t}, \qquad \mathbf{t} = C - RB. \tag{2}$$

In particular $T(B) = C$. Repeating this placement gives $\mathbf{x}_j = T^j(B)$, so $\mathbf{x}_0 = B$, $\mathbf{x}_1 = C$, and every consecutive segment has length L.

For a nonidentity rotation, let $\widehat{\mathbf{h}}$ be a directed unit vector along its screw axis and let $\theta \in [0, \pi]$ be its principal rotation angle. We define

$$d = (C - B) \cdot \widehat{\mathbf{h}} = \mathbf{t} \cdot \widehat{\mathbf{h}}, \qquad \phi = \arccos\left(\frac{d}{L}\right) \in [0, \pi]. \tag{3}$$

The equality follows because $R\widehat{\mathbf{h}} = \widehat{\mathbf{h}}$, which implies $\widehat{\mathbf{h}} \cdot RB = \widehat{\mathbf{h}} \cdot B$. The radius r is the perpendicular distance from B (or any $\mathbf{x}_j$) to the screw-axis line.

If $r > 0$, suitable perpendicular unit vectors $\mathbf{e}_1, \mathbf{e}_2$ and a point $\mathbf{a}$ on the axis give

$$\mathbf{x}_j = \mathbf{a} + r\{\cos(j\theta)\mathbf{e}_1 + \sin(j\theta)\mathbf{e}_2\} + jd\widehat{\mathbf{h}}, \qquad \mathbf{e}_2 = \widehat{\mathbf{h}} \times \mathbf{e}_1. \tag{4}$$

This is a statement about an ordered joint orbit, rather than merely the continuous cylinder or circle containing it.

2.2 The zero-twist reference

For $0 < \alpha < \pi$, define

$$\alpha = \arccos(-\mathbf{n}_B \cdot \mathbf{n}_C), \qquad \mathbf{e} = \frac{C - B}{L}, \qquad \mathbf{u} = \frac{\mathbf{n}_B \times (-\mathbf{n}_C)}{\sin\alpha}. \tag{5}$$

Let A be the shortest rotation that sends $\mathbf{n}_B$ to $-\mathbf{n}_C$. It has axis $\mathbf{u}$ and angle α. Our zero-twist placement uses this rotation. The additional twist is about the outward normal of the receiving face:

$$R = \mathrm{Rot}(\mathbf{n}_C, \tau)A. \tag{6}$$

For directed-axis formulas, first choose a representative $-\pi < \tau < \pi$. The radius and principal rotation remain periodic in τ; the directed-axis choice can reverse when the representative crosses the half-turn boundary.

This reference does not assert that $\tau = 0$ matches the vertices of two regular faces. Vertex matching depends on directions within the face plane and is derived separately in Section 8.

Proposition 2.1. *Under (1), for distinct nonopposite faces,*

$$L = 2\rho\cos(\alpha/2), \qquad \mathbf{e} = \cos(\alpha/2)\mathbf{n}_C + \sin(\alpha/2)(\mathbf{n}_C \times \mathbf{u}), \qquad \mathbf{e} \cdot \mathbf{u} = 0. \tag{7}$$

Proof. Since $\mathbf{n}_B \cdot \mathbf{n}_C = -\cos\alpha$, $\|\mathbf{n}_C - \mathbf{n}_B\|^2 = 2 + 2\cos\alpha = 4\cos^2(\alpha/2)$. This proves the first identity and gives $\mathbf{e} = (\mathbf{n}_C - \mathbf{n}_B)/(2\cos(\alpha/2))$. The vector triple-product identity gives

$$\mathbf{n}_C \times \mathbf{u} = \frac{-\mathbf{n}_B - \cos\alpha\,\mathbf{n}_C}{\sin\alpha}.$$

Substitution and the half-angle identities prove the second identity. The final identity follows because $\mathbf{u}$ is perpendicular to both normals. □

3 A common closed form for the four parameters

Theorem 3.1. *Suppose (1) holds, $0 < \alpha < \pi$, and $-\pi < \tau < \pi$. Use the placement (2) and reference rotation (6). Define*

$$Q(\alpha, \tau) = 1 - \cos^2(\alpha/2)\cos^2(\tau/2). \tag{8}$$

Then $Q > 0$, and the directed screw axis may be chosen as

$$\widehat{\mathbf{h}} = \frac{\sin(\alpha/2)\cos(\tau/2)\mathbf{u} + \sin(\tau/2)\mathbf{e}}{\sqrt{Q}}. \tag{9}$$

For this choice, the joint-centroid helix has

$$\boxed{\begin{aligned} \cos\theta &= 1-2Q, & \theta &= \arccos(1-2Q),\\ \cos\phi &= \frac{\sin(\tau/2)}{\sqrt{Q}}, & \phi &= \arccos\!\left(\frac{\sin(\tau/2)}{\sqrt{Q}}\right),\\ d &= \frac{L\sin(\tau/2)}{\sqrt{Q}}, & r &= \frac{L\sin(\alpha/2)|\cos(\tau/2)|}{2Q}. \end{aligned}} \tag{10}$$

The axis orientation agrees with the nonstraight rows of the earlier catalog: negative representative twists give $d<0$ and $\phi>\pi/2$.

Proof. Write $a=\alpha/2$ and $t=\tau/2$ temporarily. A unit quaternion for A is $(\cos a, \sin a\,\mathbf{u})$ and one for the added twist is $(\cos t, \sin t\,\mathbf{n}_C)$. Their product has scalar part $\cos a\cos t$ because $\mathbf{n}_C\cdot\mathbf{u}=0$, and vector part

$$\begin{aligned} \mathbf{v} &= \sin a\cos t\,\mathbf{u} + \sin t\{\cos a\,\mathbf{n}_C + \sin a(\mathbf{n}_C\times\mathbf{u})\}\\ &= \sin a\cos t\,\mathbf{u} + \sin t\,\mathbf{e}. \end{aligned} \tag{11}$$

The last equality is Proposition 2.1. Orthogonality gives

$$\|\mathbf{v}\|^2 = \sin^2 a\cos^2 t + \sin^2 t = 1-\cos^2 a\cos^2 t = Q.$$

Since the scalar part is nonnegative on the chosen twist interval, this is the principal-angle quaternion. Therefore

$$\sin(\theta/2)=\sqrt{Q}, \qquad \cos(\theta/2)=\cos(\alpha/2)\cos(\tau/2), \tag{12}$$

and its vector part divided by $\sqrt{Q}$ is the directed axis (9). This proves the rotation formula.

Taking the scalar product of (9) with $C-B=L\mathbf{e}$ gives $d=L\sin(\tau/2)/\sqrt{Q}$. Dividing by L proves the formula for $\cos\phi$. No choice of the position of an axis point is needed for this calculation: the axial component of the displacement is the same at every point of a rigid screw motion.

Project B, C orthogonally into a plane perpendicular to the axis. Their projected chord length c satisfies

$$c^2 = L^2 - d^2 = L^2\frac{Q-\sin^2 t}{Q} = L^2\frac{\sin^2 a\cos^2 t}{Q}. \tag{13}$$

The chord of a circle of radius r subtending θ is

$$c = 2r\sin(\theta/2). \tag{14}$$

Combining this with (12) proves the radius formula. □

The four outputs are linked by the familiar identities

$$d = L\cos\phi, \qquad r = \frac{L\sin\phi}{2\sin(\theta/2)}. \tag{15}$$

The geometric content of the theorem is the explicit connection from face data (α,τ,L) to these extrinsic helix parameters.

Corollary 3.2. *The quantities $\theta, \phi, r/L, d/L$ depend on α,τ alone. Scaling the solid by a factor $s>0$ scales L, r, d, ρ by s and leaves both angles unchanged. For nonzero twist,*

$$\tan^2\phi = \sin^2(\alpha/2)\cot^2(\tau/2). \tag{16}$$

Proof. The scale claim follows from (10). The final identity is $(L^2-d^2)/d^2$, using the two squared expressions in the proof. The squared identity does not itself specify the sign of $\cos\phi$. □

3.1 A different twist reference

Let R_0 be any other zero-twist rotation satisfying $R_0\mathbf{n}_B = -\mathbf{n}_C$. Then R_0A^{-1} fixes $\mathbf{n}_C$, so for some phase γ,

$$R_0 = \mathrm{Rot}(\mathbf{n}_C, \gamma)A, \qquad \tau_{\mathrm{eff}} = \tau + \gamma \pmod{2\pi}. \tag{17}$$

The principal rotation law is always

$$\cos(\theta/2) = \cos(\alpha/2)|\cos(\tau_{\mathrm{eff}}/2)|. \tag{18}$$

When (1) holds, replacing τ by a principal representative of τ_{eff} in Theorem 3.1 also supplies d, ϕ, r. Without the centroid-position hypothesis, the rotation law still holds but the remaining three formulas need not hold.

This is relevant to the repository convention. Its forward baseline is the product of two shortest rotations

$$R_0 = S(\mathbf{e}, \mathbf{n}_C)S(\mathbf{n}_B, -\mathbf{e}), \tag{19}$$

where $S(\mathbf{a}, \mathbf{b})$ denotes a shortest alignment. Under (1), both rotations have axis $\mathbf{u}$ and angle $\alpha/2$, because

$$\mathbf{n}_B \cdot (-\mathbf{e}) = \mathbf{e} \cdot \mathbf{n}_C = \cos(\alpha/2).$$

Their product is A. Thus $\gamma = 0$ for the Platonic face-centroid cases, and the original tabulated twists can be used directly. This conclusion is derived from the geometry, rather than selected to force numerical agreement.

4 Exact face geometry for unit edges

Throughout the remaining tables the edge length is one. For another edge length a, multiply every tabulated L, r, d by a. Define the golden ratio using a symbol distinct from the helix angle:

$$w = \sqrt{5}, \qquad g = \frac{1+w}{2}, \qquad K = \frac{\sqrt{5+2w}}{2w}, \qquad p = \sqrt{6}+\sqrt{2}, \qquad q = \sqrt{6}-\sqrt{2}. \tag{20}$$

These are fixed algebraic constants. In particular, K is the unit-edge Dodecagear radius; it is not an additional input to the theorem.

Solid	Inradius ρ	Circumradius for edge one
Tetrahedron	$\sqrt{6}/12$	$\sqrt{6}/4$
Cube	$1/2$	$\sqrt{3}/2$
Octahedron	$\sqrt{6}/6$	$1/\sqrt{2}$
Dodecahedron	$\sqrt{250+110w}/20$	$\sqrt{3}\,g/2$
Icosahedron	$g^2/(2\sqrt{3})$	$\sqrt{10+2w}/4$

Let A denote edge-adjacent faces, S, T, F successive nonopposite normal classes where needed, and O opposite faces. The labels S, T, F are defined by the scalar products in the next table, rather than by an assumed graphical ordering. Counts are relative to a fixed incoming face and exclude that face.

Solid	Class	Faces	$\cos\alpha = -\mathbf{n}_B \cdot \mathbf{n}_C$	L
Tet	A	3	$1/3$	$1/3$
Cub	A	4	0	$1/\sqrt{2}$
	O	1	1	1
Oct	A	3	$-1/3$	$\sqrt{2}/3$
	S	3	$1/3$	$2/3$
	O	1	1	$\sqrt{6}/3$
Dod	A	5	$-1/w$	g^2/w
	S	5	$1/w$	g^3/w
	O	1	1	$\sqrt{250+110w}/10$
Ico	A	3	$-w/3$	$g/3$
	S	6	$-1/3$	$g^2/3$
	T	6	$1/3$	$\sqrt{2}\,g^2/3$
	F	3	$w/3$	$g^3/3$
	O	1	1	$g^2/\sqrt{3}$

4.1 Coordinates and scalar products

Unit-edge tetrahedral vertices can be taken as $(\pm1, \pm1, \pm1)/(2\sqrt{2})$ with product of the three signs positive. The normals of distinct faces have dot product $-1/3$. Alternatively, if two faces omit vertices v_i, v_j, subtracting their centroids gives $(v_j - v_i)/3$, proving $L = 1/3$ directly.

The cube has vertices $(\pm1/2, \pm1/2, \pm1/2)$ and normals along the coordinate axes. The octahedron has vertices on those axes at distance $1/\sqrt{2}$ from the origin. Its face centers are $(\pm1, \pm1, \pm1)/(3\sqrt{2})$. Changing k signs from a reference face gives

$$L_k = \frac{\sqrt{2k}}{3}, \qquad \cos\alpha_k = \frac{2k-3}{3}, \qquad k = 1, 2, 3. \tag{21}$$

Dodecahedral unit-edge vertices can be taken as $g/2$ times

$$(\pm1, \pm1, \pm1), \quad (0, \pm g^{-1}, \pm g), \quad (\pm g^{-1}, \pm g, 0), \quad (\pm g, 0, \pm g^{-1}).$$

The face normals point to the vertices of the dual icosahedron. Dividing

$$(0, \pm1, \pm g), \quad (\pm1, \pm g, 0), \quad (\pm g, 0, \pm1)$$

by $\sqrt{1+g^2}$ gives their unit directions. Relative to one of these directions, the other eleven have dot products $1/w, -1/w, -1$ with counts $5, 5, 1$. For example, $g/(1+g^2) = 1/w$. The table then follows from (7).

Icosahedral unit-edge vertices are half the twelve coordinate vectors just listed. Their face normals are the directions of a dual dodecahedron: eight vectors $(\pm1, \pm1, \pm1)/\sqrt{3}$ and twelve vectors formed from $(0, \pm g, \pm g^{-1})/\sqrt{3}$ by cyclic coordinate permutations. Relative to $(-1, 1, 1)/\sqrt{3}$, the other nineteen have dot products

$$w/3, \quad 1/3, \quad -1/3, \quad -w/3, \quad -1$$

with counts $3, 6, 6, 3, 1$. Negating these dot products gives the listed $\cos\alpha$. Computing distances from the origin to the face planes of these coordinate solids gives the inradii above; thus the length table uses the same edge normalization as the vertex coordinates.

5 Circular, straight, and singular joint orbits

5.1 Zero twist and the circular limit

For nonopposite faces, putting $\tau = 0$ gives

$$d = 0, \qquad \phi = \pi/2, \qquad \theta = \alpha, \qquad r = \frac{L}{2\sin(\alpha/2)}. \tag{22}$$

The joint orbit lies on a circle. It closes after finitely many steps exactly when $\theta/(2\pi)$ is rational. A catalog name ending in "torus" does not itself establish finite closure of the ordered orbit or of the solid chain.

As τ runs from 0 to π, the formulas give d increasing from zero to L and r decreasing to zero. For example, writing $s = \sin(\tau/2)$,

$$\frac{d}{L} = \frac{s}{\sqrt{\sin^2(\alpha/2) + \cos^2(\alpha/2)s^2}},$$

whose derivative with respect to s is positive. Similarly, the expression for r is increasing as a function of $|\cos(\tau/2)|$. These statements describe the continuous family of joining twists, not only the matching twists selected for a catalog.

5.2 A straight orbit can hide a rotation

At $\tau = \pm\pi$ and nonopposite faces, the continuous formulas give $r = 0$, $|d| = L$, and $\theta = \pi$. The points lie on the screw axis even though the solid undergoes a half-turn. The limiting directed axes for $+\pi$ and $-\pi$ are opposite. In all straight catalog rows we choose the joint-line direction from B to C, record $d = L$, and set $\phi = 0$.

For opposite faces $\alpha = 0$, $A = I$ and $\mathbf{e} = \mathbf{n}_C$. If the twist is nonzero, the screw axis is the line through both face centers. Hence $r = 0$ and the same forward convention gives $d = L, \phi = 0$. Direct substitution into the directed-axis expression would instead give $\cos\phi = \mathrm{sgn}(\sin(\tau/2))$ for a principal nonzero twist. The difference is an axis-direction choice, not a different line of joint points. The full subunit rotation is still the principal angle of the twist.

The earlier catalog records $\theta = 0$ for each straight point sequence. Here θ always denotes the principal rotation of the specified rigid placement. The two conventions differ in precisely these five rows:

Name	Twist (deg.)	Subunit θ (deg.)	Catalog marker (deg.)
Boxbeam	-180	180	0
Staircase	-180	180	0
Octabeam	-60	60	0
Dodecabeam	-108	108	0
Pearlshaft	-60	60	0

The angular step is unidentifiable from collinear joint points alone. This does not imply that the specified motion of a labeled subunit has no rotation.

5.3 Pure translation and a repeated joint center

For opposite faces and zero twist, $Q = 0$ and the motion is pure translation. The formulas with Q in the denominator must not be used. The joint orbit is a line; we record its natural joint-line representation as $r = 0, d = L, \phi = 0, \theta = 0$. A pure translation does not select a unique axis line among parallel lines.

Selecting the same face twice gives $B = C$ and $L = 0$, so the joint orbit is stationary. The shortest normal alignment is then antipodal and its axis is not uniquely specified by the normals. Such same-face constructions are excluded here, as they are from the displayed 28-row catalog. A stationary joint orbit by itself does not establish that the entire rigid motion is the identity.

6 Length normalization in the earlier catalog

The earlier table does not state a uniform length normalization in its caption or column headings. Its entries agree with the geometries built by the associated THREE.js constructors [6]. The tetrahedron, octahedron, dodecahedron, and icosahedron constructors use radius one, meaning circumradius one. The cube constructor instead gives edge length one.

If a_0 is the original constructor's edge length, the conversion is

$$r_{\text{edge 1}} = \frac{r_{\text{catalog}}}{a_0}, \qquad d_{\text{edge 1}} = \frac{d_{\text{catalog}}}{a_0}. \tag{23}$$

Use underlying values before rounding when possible. The factors are

Solid	Constructor edge a_0	Catalog-to-unit-edge multiplier
Tet	$\sqrt{8/3}$	$\sqrt{6}/4$
Cub	1	1
Oct	$\sqrt{2}$	$1/\sqrt{2}$
Dod	$2/(\sqrt{3}\,g)$	$\sqrt{3}\,g/2$
Ico	$4/\sqrt{10+2w}$	$\sqrt{10+2w}/4$

The code first scales a constructor geometry by s to obtain a requested joint-center distance. Its factor named `SCALE_TO_UNIT_EDGE` is $1/s$, which restores the original constructor dimensions. To obtain unit edges, the factor would instead be $1/(sa_0)$. Thus the normalization issue is localized and can be corrected by scaling the length columns. It does not change θ, ϕ or the scale-independent formulas.

For instance, the tetrahedron with circumradius one has $a_0 = \sqrt{8/3}$ and $L = 2\sqrt{6}/9$. At twist $-120°$ it has

$$r = \frac{\sqrt{2}}{15}, \qquad d = -\frac{2}{\sqrt{15}},$$

which round to $0.094, -0.516$ as in the catalog. At edge length one, the same formulas instead give $r = \sqrt{3}/30$ and $d = -1/\sqrt{10}$.

Uniform edge normalization can create equal (r, d) pairs across different solids. For example, Tetratorus, Octagear, and Icosa Tree Star all have $r = \sqrt{3}/6, d = 0$ at edge length one, but different angular steps. Their ordered joint orbits should therefore not be merged using only (r, d).

7 Exact parameters for the 28 original catalog inputs

The tables in this section retain every original name, selected C-face, and twist from Table 1 of [1]. Face 0 is the incoming B face; the numerical face labels follow the pinned repository implementation. Lengths use edge length one, and w, g, K, p, q are the fixed constants in (20). All square roots denote their nonnegative real values. Angles are recovered by $\theta = \arccos(\cos\theta)$ and $\phi = \arccos(\cos\phi)$. Numeric twists in these tables are degrees.

The dagger on $\cos\phi$ denotes the forward-axis convention for straight joint rows. Their $\cos\theta$ values describe the full rigid placement, not the earlier table's zero marker. The three class-T icosahedral twists remain literal catalog inputs in this section; exact vertex-matching replacements appear in Section 8.2. Thus a successful parameter comparison is not being used as evidence of vertex matching.

7.1 Tetrahedral and cubic entries

For a tetrahedron $L = 1/3$, $\cos\alpha = 1/3$, and the radius specialization is

$$r_{\mathrm{Tet}}(\tau) = \frac{\sqrt{3}L|\cos(\tau/2)|}{2(1 + 2\sin^2(\tau/2))}. \tag{24}$$

This expression is tetrahedron-specific; the face-angle constants change for other solids. The table includes the two tetrahedral representatives.

Name	C-face	τ (deg.)	$\cos\theta$	$\cos\phi$	r	d
Tetrahelix	1	-120	$-\frac{2}{3}$	$-\frac{3}{\sqrt{10}}$	$\frac{\sqrt{3}}{30}$	$-\frac{1}{\sqrt{10}}$
Tetratorus	1	0	$\frac{1}{3}$	0	$\frac{\sqrt{3}}{6}$	0

For the cube, face 1 is opposite face 0 and face 2 is adjacent. Their lengths are 1 and $1/\sqrt{2}$, respectively.

Name	C-face	τ (deg.)	$\cos\theta$	$\cos\phi$	r	d
Boxbeam	1	-180	-1	$1^\dagger$	0	1
Staircase	2	-180	-1	$1^\dagger$	0	$\frac{1}{\sqrt{2}}$
Blockhelix	2	-90	$-\frac{1}{2}$	$-\sqrt{\frac{2}{3}}$	$\frac{\sqrt{2}}{6}$	$-\frac{1}{\sqrt{3}}$
Cubatorus	2	0	0	0	$\frac{1}{2}$	0

7.2 Octahedral entries

For edge-sharing faces $L = \sqrt{2}/3$ and $\cos\alpha = -1/3$; for faces sharing only a vertex $L = 2/3$ and $\cos\alpha = 1/3$. Consequently

$$r_{\mathrm{Oct}}(\tau) = \begin{cases} \dfrac{\sqrt{3}|\cos(\tau/2)|}{3(2+\sin^2(\tau/2))}, & A, \\ \dfrac{\sqrt{3}|\cos(\tau/2)|}{3(1+2\sin^2(\tau/2))}, & S, \\ 0, & O. \end{cases} \tag{25}$$

Repository faces $1, 3, 7$ are class A, faces $2, 4, 6$ are class S, and face 5 is opposite. The nonopposite matching twists are $0, \pm 120°$; opposite matching twists are $-60, 60, 180°$.

Name	C-face	τ (deg.)	$\cos\theta$	$\cos\phi$	r	d
Octabeam	5	-60	$\frac{1}{2}$	$1^\dagger$	0	$\frac{\sqrt{6}}{3}$
Octaspiky	1	-120	$-\frac{5}{6}$	$-\frac{3}{\sqrt{11}}$	$\frac{2\sqrt{3}}{33}$	$\frac{-\sqrt{22}}{11}$
Octamedium	2	-120	$-\frac{2}{3}$	$-\frac{3}{\sqrt{10}}$	$\frac{\sqrt{3}}{15}$	$\frac{-\sqrt{10}}{5}$
Octagear	1	0	$-\frac{1}{3}$	0	$\frac{\sqrt{3}}{6}$	0
Treestar	2	0	$\frac{1}{3}$	0	$\frac{\sqrt{3}}{3}$	0

7.3 Dodecahedral entries

Faces 1 and 2 represent classes A and S, and face 8 is opposite. The nonopposite matching twists are $0, \pm 72°, \pm 144°$. The two continuous angle functions are particularly compact:

$$\cos\phi_A(\tau) = \frac{\sqrt{10}\sin(\tau/2)}{\sqrt{10-(5-w)\cos^2(\tau/2)}},$$
$$\cos\phi_S(\tau) = \frac{\sqrt{10}\sin(\tau/2)}{\sqrt{10-(5+w)\cos^2(\tau/2)}}. \tag{26}$$

The exact radius formulas below use K, the fixed radical in (20), to avoid long rationalized denominators.

Name	C-face	τ (deg.)	$\cos\theta$	$\cos\phi$
Dodecabeam	8	-108	$\frac{1-w}{4}$	$1^\dagger$
Dodecadoubler	1	-144	$\frac{-2w-5}{10}$	$-\sqrt{\frac{10}{13-w}}$
The Alternater	2	-144	$\frac{-w-15}{20}$	$-\sqrt{\frac{10}{17-3w}}$
Dodecashaft	1	-72	$\frac{w-15}{20}$	$-\sqrt{\frac{10}{3w+17}}$
Dodecagear	1	0	$\frac{-w}{5}$	0
Dodecacorkscrew	2	-72	$\frac{2w-5}{10}$	$-\sqrt{\frac{10}{w+13}}$
Dodecadonut	2	0	$\frac{w}{5}$	0

Name	r	d
Dodecabeam	0	$\frac{\sqrt{110w+250}}{10}$
Dodecadoubler	$\frac{2Kw}{2w+15}$	$-\frac{g^2}{w}\sqrt{\frac{10}{13-w}}$
The Alternater	$\frac{4Kw}{w+35}$	$-\frac{g^3}{w}\sqrt{\frac{10}{17-3w}}$
Dodecashaft	$\frac{4Kg^2w}{35-w}$	$-\frac{g^2}{w}\sqrt{\frac{10}{3w+17}}$
Dodecagear	K	0
Dodecacorkscrew	$\frac{2Kg^2w}{15-2w}$	$-\frac{g^3}{w}\sqrt{\frac{10}{w+13}}$
Dodecadonut	Kg^2	0

For example, Dodecashaft uses $\tau = -72°$ and class A. Inserting $\sin^2 36° = (5-w)/8$ and $\cos^2 36° = (3+w)/8$ gives $\cos\phi = -\sqrt{10/(17+3w)}$. Multiplication by $L = g^2/w$ gives its tabulated advance. Such factorization is used throughout the length tables.

7.4 Icosahedral entries

The representative faces $1, 2, 8, 12, 13$ have

$$\cos\alpha = -w/3, -1/3, 1/3, w/3, 1,$$

respectively. In class T, $L = \sqrt{2}\, g^2/3$ and

$$\cos\phi_T(\tau) = \frac{\sqrt{3}\sin(\tau/2)}{\sqrt{1+2\sin^2(\tau/2)}}. \tag{27}$$

The three original twists for this class are $165°, -75°, 45°$. Here p, q abbreviate the radicals arising from the exact sine and cosine of $15°$ and $75°$. The tables describe these precise input angles, although they are not exact vertex matches.

Name	C-face	τ (deg.)	$\cos\theta$	$\cos\phi$
Pearlshaft	13	-60	$\frac{1}{2}$	$1^\dagger$
Quasi-planar	8	165	$\frac{-p-2}{6}$	$\sqrt{\frac{3\,(p+4)}{2\,(p+8)}}$
Two Strands	1	120	$\frac{-w-9}{12}$	$\sqrt{\frac{18}{w+21}}$
Slow Twist	12	120	$\frac{w-9}{12}$	$\sqrt{\frac{18}{21-w}}$
Rock Candy	2	120	$-\frac{5}{6}$	$\frac{3}{\sqrt{11}}$
Icosa Tree Star	1	0	$\frac{-w}{3}$	0
Icosacorkscrew	8	-75	$\frac{q-2}{6}$	$-\sqrt{\frac{3\,(4-q)}{2\,(8-q)}}$
Planar point cluster	2	0	$-\frac{1}{3}$	0
Big Icosacorkscrew	8	45	$\frac{-1+\sqrt{2}}{3}$	$\sqrt{\frac{9-3\sqrt{2}}{14}}$
The Wheel	12	0	$\frac{w}{3}$	0

Name	r	d
Pearlshaft	0	$\frac{g^2}{\sqrt{3}}$
Quasi-planar	$\frac{g^2\sqrt{4-p}}{\sqrt{3}\,(p+8)}$	$g^2\sqrt{\frac{p+4}{3\,(p+8)}}$
Two Strands	$\frac{2g^2}{\sqrt{3}\,(w+21)}$	$g\sqrt{\frac{2}{w+21}}$
Slow Twist	$\frac{2g^2}{\sqrt{3}\,(21-w)}$	$g^3\sqrt{\frac{2}{21-w}}$
Rock Candy	$\frac{\sqrt{6}g^2}{33}$	$\frac{g^2}{\sqrt{11}}$
Icosa Tree Star	$\frac{\sqrt{3}}{6}$	0
Icosacorkscrew	$\frac{g^2\sqrt{q+4}}{\sqrt{3}\,(8-q)}$	$-g^2\sqrt{\frac{4-q}{3\,(8-q)}}$
Planar point cluster	$\frac{g^2}{2\sqrt{6}}$	0
Big Icosacorkscrew	$\frac{g^2\sqrt{17\sqrt{2}+26}}{14\sqrt{3}}$	$g^2\sqrt{\frac{3-\sqrt{2}}{21}}$
The Wheel	$\frac{g^4}{2\sqrt{3}}$	0

8 Analytical face matching and the icosahedral correction

8.1 A face-plane calculation

Suppose the two selected faces are congruent regular m-gons. Choose a vertex P_B of the incoming face and a vertex P_C of the receiving face. After the shortest normal alignment A, put

$$\mathbf{a} = A(P_B - B), \qquad \mathbf{b} = P_C - C. \tag{28}$$

These have equal length and lie in the plane perpendicular to $\mathbf{n}_C$. The twist that aligns the selected vertices is

$$\beta = \operatorname{atan2}(\mathbf{n}_C \cdot (\mathbf{a} \times \mathbf{b}),\ \mathbf{a} \cdot \mathbf{b})\,. \tag{29}$$

Matching this one vertex matches the entire regular polygon as an unordered vertex set: all its vertices are equally spaced on the same circumcircle. Every matching twist is therefore

$$\tau = \beta + \frac{2\pi j}{m} \pmod{2\pi}, \qquad j = 0, \ldots, m-1. \tag{30}$$

Choosing a different receiving vertex changes β by a multiple of $2\pi/m$ and hence gives the same set of matching twists.

Substitution of the coordinate solids in Section 4 gives the following bases. The C-face numbering is the same as in the catalog.

Solid	Faces or class	Base twist	Increment
Tet	all three other faces	0	120°
Cub	all five other faces	0	90°
Oct	nonopposite	0	120°
	opposite, face 5	60°	120°
Dod	nonopposite	0	72°
	opposite, face 8	36°	72°
Ico	A, S, F	0	120°
	opposite, face 13	60°	120°
	T: faces $8, 11, 19$	β	120°
	T: faces $9, 10, 17$	$-\beta$	120°

The last two rows use the exact angle derived next. A base of 120° in the original implementation is equivalent to zero modulo the triangular increment; a base of 45° is not equivalent to the derived β.

8.2 An exact icosahedral base angle

Use the unit-edge icosahedral faces numbered 0 and 8. With the standard coordinates used in the verifier, their vertices are

$$F_B = \{(-g/2, 0, 1/2),\ (0, 1/2, g/2),\ (-1/2, g/2, 0)\},$$
$$F_C = \{(0, 1/2, -g/2),\ (-g/2, 0, -1/2),\ (0, -1/2, -g/2)\}.$$

Thus

$$B = \frac{g^2}{6}(-1, 1, 1), \qquad C = \frac{1}{6}(-g, 0, -g^3), \qquad \mathbf{n}_B = \frac{(-1, 1, 1)}{\sqrt{3}}, \qquad \mathbf{n}_C = \frac{(-g^{-1}, 0, -g)}{\sqrt{3}}. \tag{31}$$

The dot product is $\mathbf{n}_B \cdot \mathbf{n}_C = -1/3$, so $\cos\alpha = 1/3$. For a direct calculation of A, let $\mathbf{v} = \mathbf{n}_B \times (-\mathbf{n}_C)$ and let $[\mathbf{v}]_\times$ be its cross-product matrix. The shortest alignment is

$$A = I + [\mathbf{v}]_\times + \frac{[\mathbf{v}]_\times^2}{1 + \cos\alpha}. \tag{32}$$

Select $P_B = (-g/2, 0, 1/2)$ and $P_C = (-g/2, 0, -1/2)$ in (28). Each face has circumradius $1/\sqrt{3}$, so $\|\mathbf{a}\|^2 = \|\mathbf{b}\|^2 = 1/3$. Direct substitution gives

$$\boxed{\cos\beta = \frac{3w - 1}{8}, \qquad \sin\beta = \frac{\sqrt{3}(1 + w)}{8}, \qquad \beta \simeq 44.4775121859^\circ.} \tag{33}$$

The two displayed numbers have squared sum one and are positive, fixing the quadrant. The three matching twists for face 8 are

$$\boxed{\beta - 120^\circ, \qquad \beta, \qquad \beta + 120^\circ.} \tag{34}$$

Their sine and cosine values are especially simple:

Twist	Cosine	Sine
$\beta - 120°$	$1/4$	$-\sqrt{15}/4$
β	$(3w-1)/8$	$\sqrt{3}(1+w)/8$
$\beta + 120°$	$-(3w+1)/8$	$\sqrt{3}(w-1)/8$

For the three faces with base $-\beta$, the matching set is the negative of this set. These supply the opposite handedness of the same three joint-orbit parameter families. The other class-T face bases follow either by the face symmetries or by the same substitution in (29).

For the face-8 catalog representatives, every original twist is greater than its matching replacement by $45° - \beta \simeq 0.5224878141°$. The nearest corresponding vertices then differ by the positive distance

$$\frac{2}{\sqrt{3}} \sin\left(\frac{45° - \beta}{2}\right) \simeq 0.00526492 \tag{35}$$

at edge length one. This is a finite geometric mismatch, not floating-point roundoff. The implementation's use of the same positive 45° base for all six class-T faces also misses the negative-base family. Nevertheless, all three literal catalog configurations remain well-defined normal-aligned joint-centroid helices. Their parameter formulas in Section 7.4 are valid independently of exact vertex matching.

8.3 Exact corrected parameters

Retaining the names as labels for their corrected joining rules gives the following replacements. The other 25 catalog entries are unchanged by this vertex-matching correction. Edge length is one.

Name	τ	$\cos\theta$	$\cos\phi$
Quasi-planar	$\beta + 120°$	$\dfrac{-3w-5}{12}$	$\dfrac{3g}{\sqrt{3w+17}}$
Icosacorkscrew	$\beta - 120°$	$-\dfrac{1}{6}$	$-\dfrac{3}{\sqrt{14}}$
Big Icosacorkscrew	β	$\dfrac{3w-5}{12}$	$\dfrac{3}{g\sqrt{17-3w}}$

Name	r	d
Quasi-planar	$\dfrac{2}{\sqrt{3}\,(3w+17)}$	$g^3\sqrt{\dfrac{2}{3w+17}}$
Icosacorkscrew	$\dfrac{\sqrt{15}g^2}{21}$	$\dfrac{-g^2}{\sqrt{7}}$
Big Icosacorkscrew	$\dfrac{2g^4}{\sqrt{3}\,(17-3w)}$	$g\sqrt{\dfrac{2}{17-3w}}$

Name	τ (deg.)	r	d	θ (deg.)	ϕ (deg.)
Quasi-planar	164.477512	0.048705	1.230350	167.3389	4.4993
Icosacorkscrew	−75.522488	0.482838	−0.989524	99.5941	143.3008
Big Icosacorkscrew	44.477512	0.769004	0.713275	81.8161	54.6937

The source analog counts for these three rows are six each: three faces with base β and three with base $-\beta$, paired by handedness. There are

$$3 \cdot 3 + 5 \cdot 4 + 7 \cdot 3 + 11 \cdot 5 + 19 \cdot 3 = 162$$

distinct-face matching constructions across the five solids. Their counts by representative, retaining the earlier names, are

Solid	Representative multiplicities in catalog order	Total
Tet	$6, 3$	9
Cub	$4, 4, 8, 4$	20
Oct	$3, 6, 6, 3, 3$	21
Dod	$5, 10, 10, 10, 5, 10, 5$	55
Ico	$3, 6, 6, 6, 12, 3, 6, 6, 6, 3$	57

These multiplicities group the fixed-solid, fixed-edge joint-centroid orbits, with handedness paired and with the original catalog's grouping of straight orbits retained. Different twists about an opposite face can give the same straight joint orbit while rotating the solid by different angles. The multiplicities do not identify different solids or different ordered circular orbits just because their containing circles happen to agree.

9 Verification, scope, and further work

The supplemental exact catalog records both the literal 28-row input set and the three matching corrections. A symbolic generator checks every displayed $r^2, d^2, \cos^2\phi$ against the theorem, with the signs checked separately, and obtains $\cos\theta$ directly. These checks reduce identities in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ after clearing denominators; they do not rely on agreement of decimal expansions. The exact vertex calculation of $\cos\beta, \sin\beta$ is checked symbolically as well.

An independent standard-library Python program constructs the unit-edge coordinate solids and their numbered faces, obtains rotations from Rodrigues matrices, obtains axes from their antisymmetric parts in nondegenerate cases, and measures radius using projected three-point circumcircles. It compares the exact expressions with those constructions, converts the lengths back to the original constructor scale, and checks all four printed catalog columns. The five zero-angle markers are converted to the stated straight-point convention for that comparison.

In the accompanying run, all 28 rows agree with the original table at its three-decimal precision. The largest discrepancies against the rounded entries are less than 0.000497 in radius, 0.000453 in advance, 0.000476° in rotation, and 0.000444° in segment-axis angle. The exact expressions and the independent numerical constructions agree within 5×10^{-16} for their algebraic outputs.

The vertex audit checks every corner under every placement. All 162 corrected distinct-face rules match to numerical precision; 18 rules from the original implementation fail that check, precisely the three choices on each of the six class-T icosahedral faces. The maximum corrected unit-edge vertex discrepancy is less than 6×10^{-16} in the accompanying run. A further 3,948 checks cover all ordered nonparallel face pairs of the five solids at seven continuous twist values. Their largest parameter discrepancy is less than 2×10^{-15}.

Numerical checks support the derivations and catch transcription errors; they do not replace the geometric and algebraic proofs. Vertex matching also does not establish the absence of collisions

between nonadjacent members of an infinite chain. The present results concern the prescribed rigid placement and the orbit of its joint centers.

The equal-radius centroid hypothesis is the useful boundary of the present position formulas. It permits generalization beyond the Platonic solids whenever two selected joint centers lie at equal distances on their face normals from a common point. Extending r, d, ϕ to arbitrary joint positions requires the translational part of the placement in addition to the two normals. The rotational half-angle law already survives that extension. Further work can address collision-free realizations, exact closure conditions, and geometrically appropriate illustrations of these relationships.

A Numerical unit-edge crosswalk

This table evaluates the original input twists, including the three literal icosahedral choices. It is a crosswalk for reviewing the radical tables; the exact expressions, rather than these rounded numbers, define the results. Here θ is the full subunit rotation and straight rows use $d > 0, \phi = 0$.

Name	τ (deg.)	r	d	θ (deg.)	ϕ (deg.)
Tetrahelix	−120	0.057735	−0.316228	131.8103	161.5651
Tetratorus	0	0.288675	0.000000	70.5288	90.0000
Boxbeam	−180	0.000000	1.000000	180.0000	0.0000
Staircase	−180	0.000000	0.707107	180.0000	0.0000
Blockhelix	−90	0.235702	−0.577350	120.0000	144.7356
Cubatorus	0	0.500000	0.000000	90.0000	90.0000
Octabeam	−60	0.000000	0.816497	60.0000	0.0000
Octaspiky	−120	0.104973	−0.426401	146.4427	154.7606
Octamedium	−120	0.115470	−0.632456	131.8103	161.5651
Octagear	0	0.288675	0.000000	109.4712	90.0000
Treestar	0	0.577350	0.000000	70.5288	90.0000
Dodecabeam	−108	0.000000	2.227033	108.0000	0.0000
Dodecadoubler	−144	0.158056	−1.128508	161.3006	164.5496
The Alternater	−144	0.165307	−1.867378	149.5197	170.3063
Dodecashaft	−72	0.491851	−0.760398	129.6575	130.5007
Dodecagear	0	0.688191	0.000000	116.5651	90.0000
Dodecacorkscrew	−72	0.765348	−1.534764	93.0258	144.1102
Dodecadonut	0	1.801707	0.000000	63.4349	90.0000
Pearlshaft	−60	0.000000	1.511523	60.0000	0.0000
Quasi-planar	165	0.047037	1.230603	167.7643	4.3467
Two Strands	120	0.130101	0.474702	159.4456	28.3400
Slow Twist	120	0.161109	1.382980	124.3094	11.6407
Rock Candy	120	0.194329	0.789367	146.4427	25.2394
Icosa Tree Star	0	0.288675	0.000000	138.1897	90.0000
Icosacorkscrew	−75	0.486993	−0.986177	99.2526	143.0416
Planar point cluster	0	0.534404	0.000000	109.4712	90.0000
Big Icosacorkscrew	45	0.763752	0.719429	82.0638	54.3429
The Wheel	0	1.978609	0.000000	41.8103	90.0000

B Reproducibility and draft provenance

The catalog inputs are from Table 1 of [1]. The implementation audit uses repository revision

`8be3839de2204b966c8aedb7c28f6a6f5c18076e`.

Here `main.js` supplies the constructor scales, face numbering, and twist maps, and `js/segment_helix_math.js` supplies the adjoining construction [6]. The supplement reconstructs the numbered faces on exact coordinate solids; the independent numerical checks use those radicals evaluated in double precision.

The package contains the following material:

- `PlatonicHelicesClosedForms.tex`: self-contained LaTeX source.
- `catalog_data.py`: catalog transcription, exact-coordinate constructions evaluated numerically, and numbered faces.
- `ExactCatalog.json`: exact expressions, fixed constants, source entries, and corrected entries.
- `verify_closed_forms.py`: independent numerical checks, requiring only Python's standard library.
- `build_exact_catalog.py`: optional symbolic regeneration using SymPy; generated table fragments are also supplied.
- `verification-results.txt`: output of the accompanying run.

The manuscript source embeds its tables and has no external image or bibliography dependencies. It can be compiled with `pdflatex` twice or with `latexmk -pdf`. The symbolic supplement is optional for compilation and for running the independent verifier.

This author-review draft was developed with ChatGPT assistance in derivation, calculation, verification code, and writing. It has not been submitted or peer reviewed. No priority claim is made for the quaternion or screw-geometric ingredients.

References

[1] R. L. Read, Calculating the segmented helix formed by repetitions of identical subunits thereby generating a zoo of Platonic helices, *Mathematics* **10** (2022), 2533. doi:10.3390/math10142533.

[2] E. A. Lord, Helical structures: the geometry of protein helices and nanotubes, *Structural Chemistry* **13** (2002), 305–314. doi:10.1023/A:1015863923728.

[3] Z. Dauter and M. Jaskolski, On the helical arrangements of protein molecules, *Protein Science* **27** (2018), 643–652. doi:10.1002/pro.3356.

[4] G. Sadler, F. Fang, R. Clawson, and K. Irwin, Periodic modification of the Boerdijk–Coxeter helix (tetrahelix), *Mathematics* **7** (2019), 1001. doi:10.3390/math7101001.

[5] S. Onaka, Periodic arrangements of tetrahedra having appearances similar to that of the Boerdijk–Coxeter helix, *Scientific Reports* **14** (2024), 18260. doi:10.1038/s41598-024-69108-w.

[6] Public Invention, *Segmented Helices*, source repository, `PubInv/segmented-helixes`. Audit revision: `8be3839de2204b966c8aedb7c28f6a6f5c18076e`.